Round 14 - Response-Cashout Moment Propagator (RCMP)

One-use cumulants, long-use first-two-moment response

[Derived design] Evaluate every newborn third-cumulant source exactly once, at age 1, while its legs are simplest. Convert that source immediately through the existing Edgeworth closure into a dense post-ReLU mean/covariance response, then discard the cubic source. All later memory is carried by the corrected mean m and full covariance C. No old K3 sources, no basis join, no old tier. Exact algebraic core: 91 dense 1024x1024 products per network = 91 u with ordinary GEMM (utilization 0.08887 before O(n^2) closure work).

1. Why this design, not another K3 representation

[Quoted] V29 is raw 2.13e-8 at utilization 0.254; the engineered chain is raw 2.16e-8 at 0.189; leaders are raw 1.54-1.61e-8 at 0.125-0.139. The channel oracles give variance x0.53, full K3 input x0.61, K4 diagonal x0.70, and true mean at every layer removes about 80% of final error.

[Quoted from the earlier K3m experiment] About 86% of the old-source effect reaches the output through the next one to two mean shifts, mainly via covariance/variance -> E[ReLU].

[Inference] That observation changes the target. The expensive object is not valuable because the final answer needs a cubic tensor; it is valuable because the tensor perturbs the first two moments at later ReLUs. RCMP cashes out that response at the first use, while the source is exact and uncompressed, and stores the response in C. The remaining approximation is therefore the roughly 14% direct higher-order path, not a 4-7% D21 approximation accumulated over many transported sources.

2. State and per-layer update

[Derived] State after layer l: m_l in R^n, C_l in R^(n x n), and one pending age-1 packet for layer l+1: D21^pend in R^(n x n), d3^pend in R^n, plus the current memoryless K4 core quantities already used by V29. The pending packet exists for exactly one layer and is never recursively transported.

A. Linear map to the next pre-activation. For z_l = h_(l-1) W_(l-1):

$$\mu_l = m_{(l-1)} W_{(l-1)}, \qquad \Sigma_l = W_{(l-1)}^{\top} C_{(l-1)} W_{(l-1)}.$$

B. Closure with a one-use packet. Let F_G be the exact Gaussian univariate/bivariate ReLU moment map and F_E be the existing V29 Edgeworth closure. Apply only the pending higher-cumulant packet:

$$\Delta m_l = F_E^{\text{mean}}(\mu_l, \Sigma_l, d3^{\text{pend}}, d4^{\text{pend}}) - F_G^{\text{mean}}(\mu_l, \Sigma_l)$$

$$\Delta C_l = F_E^{\text{cov}}(\mu_l, \Sigma_l, D21^{\text{pend}}, K4^{\text{pend}}) - F_G^{\text{cov}}(\mu_l, \Sigma_l)$$

$$m_l = F_G^{\text{mean}} + \Delta m_l, \qquad C_l = F_G^{\text{cov}} + \Delta C_l.$$

[Derived] Once Delta m_l and Delta C_l are added, the source is forgotten. Its later effect is carried through the ordinary dense m,C recursion. This is not a short-horizon K3 carrier: the response persists for all remaining layers.

C. Birth of the next packet. At the same ReLU, form the same newborn star source as V29 from the current moments. Using the prompt's notation,

$$a_l = \text{diag}(\Phi_l)\, C_{\text{off},l}, \qquad T_l^{\text{birth}} = \text{Sym}_3\Big(\sum_r e_r \otimes e_r \otimes a_{(l,r)}\Big) \quad \text{[with V29 birth weights]}.$$

Do not store T_l. Project it immediately through the known next map M_l. For age 1, P=M_l is already known; form A=M_l a_l and evaluate the exact V29 source slice:

$$D21_{(l+1)}^{\text{pend}} = [(P \circ P)\, \text{diag}(w_l)\, A^{\top} + 2(P \circ A)\, \text{diag}(w_l)\, P^{\top}] / 3.$$

[Derived] d3_(l+1)^pend is the diagonal/source-diagonal readout from the same age-1 source. The current memoryless K4 core G_l = diag(g_l) + lambda_l C_off,l is projected one step exactly as in the present chain. No K3 or K4 source survives beyond this packet.

3. Exact dense-product ledger at n = 1024

stage	real shape	count/network	classical cost
Layer-1 covariance Gram	(1024x1024)^T (1024x1024)	1	1 u
Layers 2-16 covariance transport	W^T C W: two 1024^3 products	30	30 u
Age-1 A formation	M_l a_l, 1024^3	15	15 u
Age-1 D21 contractions	two 1024^3 products	30	30 u
Memoryless K4 diagonal/core projection	one 1024^3 product	15	15 u
TOTAL square products	all 1024x1024 by 1024x1024	91	91 u

[Derived] FLOPs. 91 x 2^31 = 195,421,011,968 FLOPs, so square-product utilization is 91/1024 = 0.088867. Mean transports and all closure/birth elementwise work are O(n^2), not n^3.

[Estimated] Total prototype utilization. Approximately 0.09-0.10 after the O(n^2) closure work, if implemented with ordinary GEMM. This is deliberately not Strassen-first: ordinary GEMMs keep the residual-operation count low and the 91-u core is already below the 0.1 score floor.

[Estimated] Metered operations. Roughly 2,000-5,000 per network with batched closures and ordinary matmuls, comfortably below the 18,000 cap. Only if measured utilization rises above 0.10 should selected covariance products be switched to L5 Strassen.

4. Ceiling and why the key component can beat V29

- **[Quoted]** Perfect variance gives a final-MSE ratio 0.53. RCMP's primary state is the full dense covariance, so this is its largest directly accessible oracle ceiling.
- **[Quoted]** Perfect full K3 input gives 0.61 / 0.71 on the two measured networks. RCMP does not reproduce K3; it tries to reproduce the dominant effect K3 has on m and C.
- **[Quoted + inference]** The earlier response measurement assigns about 86% of old-source influence to the next one or two mean shifts. If that fraction generalizes, cashing the source into m,C is aimed at the dominant pathway rather than the tensor itself.
- **[Derived]** The age-1 source is evaluated before transport, shared-basis projection, or old-tier approximation. Therefore the local D21 packet has none of the long-source rank/basis error that V29 accumulates.
- **[Derived]** Parity is preserved: the packet is computed from the realized source and realized next W. After cashout, C carries that quenched sign/orientation; RCMP never predicts the missing content from an even ensemble statistic.

- **[Estimated]** A mature version that captures roughly 65-80% of the variance/K3-response opportunity would plausibly reduce V29 raw MSE by about 15-30%, enough to enter the 1.6-1.9e-8 regime. This is a projection, not a measured result.

5. Staged two-week evaluation plan

stage	implementation	pass/fail gate	purpose
0 - cheap falsifier	Teacher-seed post-L11 (m,C) on each network; erase K3/K4 history; run RCMP L12-L16 using only newly generated one-use packets.	PASS if the geometric-mean relative preactivation-variance error over L14-L16 is <= 0.85 x V29 on BOTH networks. Otherwise kill RCMP.	Tests whether first-two-moment cashout can actually carry the information that matters.
1 - first full prototype	Run from layer 1 with classical GEMMs, V29 newborn-star formula, V29 memoryless K4, no tuning beyond existing constants.	Deep variance error <= 0.95 x V29 and raw final MSE <= 2.35e-8 on the deterministic harness; <= 100 u; residual-safe.	A new design need not beat the tuned chain immediately; it must beat V29's component or show a much better cost/accuracy slope.
2 - analytic maturation	Use exact incremental closure differences, preserve all decisive/inactive hubs in birth, compact only proven pass-through rows/columns, validate float32/f64-sensitive terms.	Raw <= 2.05e-8 at <= 0.10-0.11 utilization; no network > residual cap.	Shows the response state is not merely a cheap but inaccurate approximation.
3 - engineering week	Profile metered operations; keep ordinary GEMM if under floor. Add L5 only to the few products needed to recover headroom; reuse closure tables and batch elementwise families.	100-network paired run: no failures, < 18k ops/network, stable score.	Turns the mathematical prototype into a grader-safe estimator.

[Estimated two-week maturity projection, conditional on Stage 0 passing strongly] raw MSE 1.65-1.85e-8 at utilization about 0.09-0.11, corresponding to an adjusted score about 1.65-2.04e-9. The lower end requires the response-state hypothesis to recover most of the variance/K3 oracle opportunity; the upper end is the more conservative target. These are engineering projections, not leaderboard predictions.

6. ONE cheap first falsifier

Teacher-seeded L11 -> L16 cashout test. Use the true post-ReLU mean and covariance at layer 11 as the only carried history. Set all old K3/K4 state to zero. From layer 12 onward run RCMP freely: Gaussian/bivariate first-two-moment propagation, plus the exact V29 newborn age-1 packet generated at each ReLU and cashed out once at the next closure. Compare preactivation variance to the stored teacher at L14, L15 and L16. Primary gate: on EACH of MLP #0 and the synthetic network, geometric-mean relative variance RMS must be <= 0.85 times the corresponding V29 error. This is intentionally a favorable seed: if RCMP cannot beat V29's variance by 15% even after all pre-L11 history has already been summarized into the TRUE (m,C), then the 'one-use cumulant -> long-use response' representation does not have enough information and should be killed before a full implementation.

7. Interpretation if it passes

[Inference] A pass would provide a coherent explanation of the leaderboard geometry: the public chain spends most of its budget carrying a tensor whose dominant downstream role is to perturb m and C, while a response estimator computes that perturbation once and then propagates only first-two-moment state. That simultaneously removes the young-source, join and old-tier costs and can improve accuracy because the source is consumed before its all-distinct content is approximated by long transport/compression.

External cross-check: the public 504aldo write-up reports V29 at raw 2.13e-8 and about 0.253 budget and says its accuracy plateaued before later versions focused on cost; the official starter kit confirms Phase-2 width 1024, depth 16 and budget 2^41 FLOPs. Those public facts are consistent with, but do not prove, the architectural hypothesis above.